

JEE MAIN REVISION MASTER: MOTION IN A PLANE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 90 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Motion In A Plane - Core Application	90 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Motion In A Plane - Core Application (Very High Yield — 90 PYQs)

Core Principle: Key problem pattern in Motion In A Plane.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Two identical bodies, projected with the same speed at two different angles cover the same horizontal range R . If the time of flight of t_1 is T_1 and of t_2 is T_2 , then $T_1 = T_2$ is true only if..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "A boy throws a ball into air at 45° from the horizontal to land it on a roof of a building of height H . If the ball attains maximum height $2H$ above the roof, then the angle of projection is..." [Ans: D]
- [2026 Evening Shift] Q#2 (MCQ): "At $t=0$, a body of mass 100 g starts moving under the influence of a force $(5i + 10j)\text{ N}$. After 2 s its position is $(2x\hat{i} + y\hat{j})\text{ m}$. Then x and y are..." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Two identical bodies, projected with the same speed at two different angles cover the same horizontal range R . If the time of flight of these bodies are 5 s and 10 s, respectively, then the value of R is ____ m. (Take $g=10\text{m/s}^2$)

- (A) 250 **[Correct]**
- (B) 25
- (C) 500
- (D) 125

Correct Answer: A

Solution Analysis: For two projectiles launched with the same initial speed u to have the same horizontal range R , their angles of projection must be complementary. Let the first angle be θ . Then the second angle must be $(90^\circ - \theta)$. The time taken for the projectile to return to the same horizontal level. $T = (2u \sin \theta / g)$... (i) And the total horizontal distance covered is the range of projectile, R . $R = \frac{u^2 \sin(2\theta)}{g} = \frac{2u^2 \sin \theta \cos \theta}{g} \Rightarrow R = \frac{u^2 \sin \theta \cos \theta}{g}$ (ii) For the first body ($T_1 = 5\text{s}$): $T_1 = \frac{2u \sin \theta}{g}$